LECTURE PLAN

Institute/Department	UNIVERSITY INSTITUTE OF ENGINEERING (UIE)	Program	Bachelor of Engineering (Information Technology)(IT201)
Master Subject Coordinator Name:	Parveen Kumar Saini	Master Subject Coordinator E-Code:	E13339
Course Name	Principles of Artificial Intelligence	Course Code	23ITT-306

Lecture	Tutorial	Practical	Self Study	Skilling	TC	TGT	TGP	Studio	Credit	Subject Type
2	0	0	0	0	0	0	0	0	2.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Minor Core	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1: To provide practical knowledge using state-of-the-art technological support for the experiential learning of our students. M2: To provide industry recommended curriculum and transparent assessment for quality learning experiences. M3: To create global linkages for interdisciplinary collaborative learning and research. M4: To nurture advanced learning platform for research and innovation for students' profound future growth. M5: To inculcate leadership qualities and strong ethical values through value based education.
Vision of the Department	To be recognized as a leading Computer Science and Engineering department through effective teaching practices and excellence in research and innovation for creating competent professionals with ethics, values and entrepreneurial attitude to deliver service to society and to meet the current industry standards at the global level.

Program Educational Objectives(PEOs)	
PEO1	Engage in successful careers in industry, academia, and public service, by applying the acquired knowledge of Science, Mathematics and Engineering, providing technical leadership for their business, profession and community.
PEO2	Establish themselves as entrepreneur, work in research and development organization and pursue higher education.
PEO3	Exhibit commitment and engage in lifelong learning for enhancing their professional and personal capabilities.

Program Specific OutComes(PSOs)	
PSO1	Exhibit attitude for continuous learning and deliver efficient solutions for emerging challenges in the computation domain.
PSO2	Apply standard software engineering principles to develop viable solutions for Information Technology Enabled Services (ITES).

Program OutComes(POs)	
PO1	Disciplinary knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.
PO2	Complex problem solving: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO3	Critical Thinking
PO4	Creativity
PO5	Communication skills: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO6	Analytical reasoning/thinking: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO7	Research related skills: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO8	Coordinating/Collaborating with others: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO9	Leadership qualities: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO10	Learning how to learn skills: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PO11	Digital and technological skills: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO12	Multicultural Competence and Inclusive Spirit.
PO13	Value inculcation: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO14	Autonomy, responsibility and accountability: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO15	Environmental awareness and action: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.
PO16	Community Engagement & Services.
PO17	Empathy

Text Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	1. Kevin Knight Elaine Rich B Nair Artificial Intelligence Tata Mcgraw Hill	Elaine Rich	Third Edition	Tata McGraw-Hill	1 July 2017
2	Artificial Intelligence for Robotics: Build intelligent robots that perform human tasks using AI tec	Francis X Govers (Author)	Newer edition	Wiley	30 August 2018
3	Artificial Intelligence: A Modern Approach by Stuart Russell	Peter Norvig	3	Pearson	1 January 2015

Reference Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	2. Artificial Intelligence: A Modern Approach, Global Edition by Russell	Russell	4	Pearson	20 May 2021
2	Artificial Intelligence: A Modern Approach	4eby Russell/Norvig	4	Publisher: Pearson	31 May 2022

Course OutCome

SrNo	OutCome
CO1	Students will be able to Understand the foundational concepts, history, applications, and terminology of Artificial Intelligence across symbolic logic (propositional & predicate calculus), search-base
CO2	Students will be able to Apply AI techniques—including logical inference, uninformed and informed search strategies, and basic neural network learning methods—to model and solve standard AI problems.
CO3	Students will be able to Analyze different state-space representations, reasoning processes (forward/backward), heuristic evaluations, and learning paradigms (supervised, unsupervised, reinforcement)
CO4	Students will be able to Evaluate the effectiveness of AI methods such as predicate logic reasoning, heuristic & optimization-based search algorithms, and training techniques (GD, SGD, backpropagation

CO5	Students will be able to Develop integrated AI solutions by designing and justifying appropriate combinations of symbolic reasoning, search strategies, and neural network architectures (MLP, CNN, RNN,

Lecture Plan Preview-Theory

Unit No	SessionNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	CO - BT Level Mapping
1	1	Chapter 1.1 (Introduction to AI)	Introduction to Artificial Intelligence – History and Evolution	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
1	2	Chapter 1.1 (Introduction to AI)	Applications of AI in the Modern World	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
1	3	Chapter 1.1 (Introduction to AI)	Attitudes Toward Intelligence, Knowledge, Human Artifice	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
1	4	Chapter 1.2 (Overview of AI Techniques & its Applications)	AI – Attempted Definitions	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
1	5	Chapter 1.2 (Overview of AI Techniques & its Applications)	Foundations of AI – Philosophy, Mathematics, Cognitive Science	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
1	7	Chapter 1.2 (Overview of AI Techniques & its Applications)	Problems in AI – Representation, Search, Uncertainty	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3
1	8	Chapter 1.3 (THE PREDICATE CALCULUS)	Approaches to AI – Symbolic, Connectionist, Evolutionary	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3
1	9	Chapter 1.3 (THE PREDICATE CALCULUS)	Propositional Calculus – Syntax, Semantics, Reasoning	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Video Lecture	CO2

1	10	Chapter 1.3 (THE PREDICATE CALCULUS)	Predicate Calculus & Inference Rules	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Video Lecture	CO2
2	6	Chapter 1.2 (Overview of AI Techniques & its Applications)	Scope of AI – Opportunities & Limitations	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
2	11	Chapter 2.1 (Problem-solving through Search)	Uninformed Search – DFS, BFS, UCS	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO2
2	12	Chapter 2.1 (Problem-solving through Search)	Informed Search – Heuristics, Best First	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO2
2	13	Chapter 2.1 (Problem-solving through Search)	Local Search & Optimization	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO4
2	14	Chapter 2.1 (Problem-solving through Search)	Partial Observations, CSP Introduction	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3
2	15	Chapter 2.1 (Problem-solving through Search)	Constraint Propagation, Backtracking & Game Playing	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3
2	16	Chapter 2.2 (Problem Solving)	State Space, Blind vs Heuristic Search & Problem Reduction	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
2	17	Chapter 2.2 (Problem Solving)	A & A* Algorithms with Sample Applications	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO2
2	18	Chapter 2.2 (Problem Solving)	Forward & Backward Reasoning	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3

2	19	Chapter 2.3 (Structures And Strategies For State Space Searches)	Graph Theory & State Space Representations	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
2	20	Chapter 2.3 (Structures And Strategies For State Space Searches)	Data-Driven vs Goal-Driven, DFS, BFS & Heuristic Searches	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO2
3	21	Chapter 3.1 (Fundamentals of Neural Networks)	Intro to Neural Networks & Biological Inspiration	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO1
3	22	Chapter 3.1 (Fundamentals of Neural Networks)	Perceptron & Activation Functions	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO2
3	23	Chapter 3.2 (Learning methods)	Learning Paradigms – Supervised, Unsupervised, RL	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3
3	24	Chapter 3.2 (Learning methods)	Hebbian & Competitive Learning	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO3
3	25	Chapter 3.2 (Learning methods)	Forward Prop, Backprop, GD & SGD	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO4
3	26	Chapter 3.2 (Learning methods)	Loss Functions, Metrics, Over/Underfitting	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO4
3	27	Chapter 3.3 (Neural Network Architectures)	Feedforward NN – Single Layer, MLP, DNN	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO2
3	28	Chapter 3.3 (Neural Network Architectures)	Convolutional Neural Networks (CNN)	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO5
3	29	Chapter 3.3 (Neural Network Architectures)	RNN, BRNN & LSTM	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO5
3	30	Chapter 3.3 (Neural Network Architectures)	Applications of Neural Networks	,T-1. Kevin Knight Elaine Rich B ,T-Artificial Intelligence for Ro,T-Artificial Intelligence: A Mod,R-2. Artificial Intelligence: A ,R-Artificial Intelligence: A Mod	PPT,Video Lecture	CO5

Assessment Model			
Sr No	Exam Name	Max Marks	Weighted Marks
1	External Theory	60	60
2	Assignment/PBL	10	10
3	Attendance Marks	2	2
4	Mid-Semester Test-1	20	10
5	Quiz	4	4
6	Surprise Test	12	4
7	Mid-Semester Test-2	20	10

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO13	PO14	PO15
CO1	2	2	2	3	3	2	2	3	2	2	2	1	2	3	1
CO2	3	3	2	2	2	2	2	1	NA	2	3	NA	NA	2	1
CO3	2	2	3	3	2	3	3	2	2	2	3	NA	1	2	1
CO4	2	3	3	2	2	2	2	2	3	2	2	1	2	2	3
CO5	2	2	2	3	3	3	3	2	2	3	2	1	3	2	2
Target	2.2	2.4	2.4	2.6	2.4	2.4	2.4	2	2.25	2.2	2.4	1	2	2.2	1.6

PO16	PO17	PSO1	PSO2
2	2	2	3
NA	NA	3	2
NA	NA	3	2
2	3	2	3
2	2	3	2
2	2.33	2.6	2.4